Bassez 2021 Breast — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 41 samples: same-sample neighbour fraction 0.37 → 0.21, iLISI median 5.51 → 9.10 of 41. One connected component (45 cells / 27 specks removed). Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

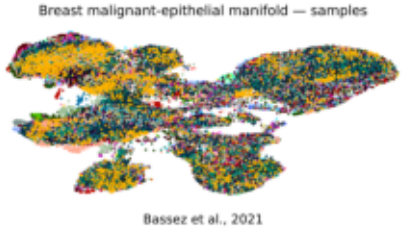
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=−0.21), both oppose Crest (pSom↔Crest r=−0.82, Gast↔Crest r=−0.24). Along pseudotime Gast/pSom fall (ρ −0.11/−0.32), Crest rises (ρ +0.32) to the terminal lobe. Joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

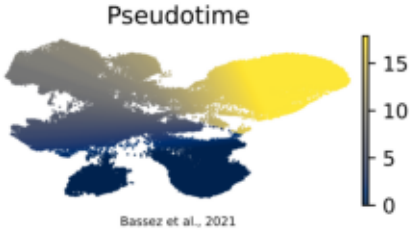
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

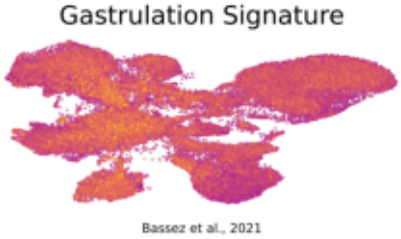
Manifold — samples



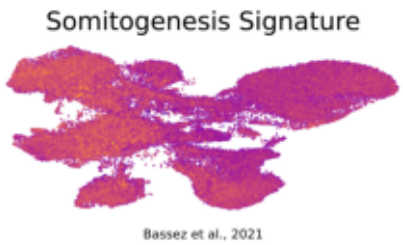
Pseudotime



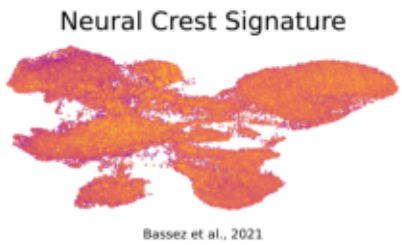
Gastrulation



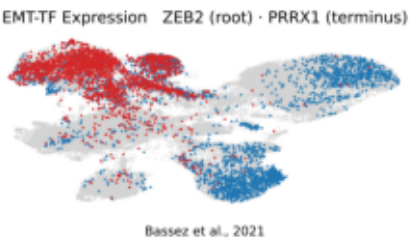
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast tumours (Bassez et al., 2021), 10x — 65,837 malignant cells / 41 samples; single low-cycling body (2% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root) → PRRX1 (terminus).

Parameter choices

Sample threshold	geom-mean of 92→3 cliff = 16 (drops 3-cell BIOKEY_34) → 41/42 samples	**Overdispersed genes**	2,498 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing flat across sweep)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 45 cells / 27 islands removed → 65,837
Geometry	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.53	**Root**	tip 2 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.014 · pSom +0.022 · Crest −0.005 (Linear-pSom/Crest)	**EMT-TFs shown**	ZEB2 (root, ρ −0.07) ↔ PRRX1 (terminus, ρ +0.07); detection-gated